Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/13/2026
PI Name:	Basner, Mathias M.D., Ph.D.		
Project Title:	Long-Term Brain Structural and Functional Consequences of Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H) (2) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS) (3) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	basner@pennmedicine.upenn.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	215-573-5866
Organization Name:	University of Pennsylvania		
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PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	03/15/2021	**End Date:**	09/30/2033
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2033 per NSSC documentation (Ed., 6/6/23). NOTE: End date changed to 03/31/2029 per L. Juliette/JSC (Ed., 5/3/22).		
Key Personnel Changes/Previous PI:	Co-I Christopher W. Jones, PhD left the University of Pennsylvania for another institution on 9/30/2025. PI Dr. Basner will assume his duties until a replacement for Dr. Jones has been identified.		

COI Name (Institution):	Gunga, Hanns-Christian M.D. (Charite - Universitatsmedizin Berlin, Germany) Gur, Ruben Ph.D. (The Trustees of the University of Pennsylvania) Hartley, Tom Ph.D. (University of York, United Kingdom) Kuehn, Simone Ph.D. (Max Planck Institute for Human Development, Berlin, Germany) Riecke, Bernhard Ph.D. (Simon Fraser University, Canada) Roalf, David Ph.D. (University of Pennsylvania) Bell, Suzanne Ph.D. (NASA Johnson Space Center) Stangl, Matthias Ph.D. (University of California, Los Angeles) Wolbers, Thomas Ph.D. (German Center for Neurodegenerative Diseases, Germany) Stahn, Alexander Ph.D. (Charite - Universitatsmedizin Berlin, Germany)
Grant/Contract No.:	80NSSC21K1698
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Spaceflight-associated neuro-ocular syndrome (SANS) describes structural and functional alterations of the eye that develop in many astronauts during long-duration spaceflight missions (>6 month). Understanding the consequences of the structural and functional changes to the eye and brain that endure long after return from spaceflight is essential for future exploration-class spaceflight missions. The primary purpose of this project is to characterize the structural and functional changes to the eye and brain, including cognitive changes, in crewmembers who have participated in past missions, calling them back several years after their most recent flights. Outcomes from astronauts who had previously participated in ≥6-month missions (n=20) will be compared to control subjects that do not have standard duration spaceflight experience (n=20). In addition to characterizing the effects observed for individual variables, all data collected during the study will be analyzed using an integrative approach to identify relationships across ocular measures, brain measures, individual crewmembers, and time. In collaboration with the integrated project team, ocular structure and function will be evaluated using magnetic resonance imaging (MRI) and 3D structural analysis, optical coherence tomography (OCT), angiography, electroretinography (ERG), visual field analysis, pneumotonometry (intraocular pressure), visual field analysis, optical biometry. MRI will be used to assess brain structural changes, and functional MRI (fMRI) and a series of cognitive batteries will be used to evaluate brain function. To better understand the neurophysiological mechanisms underlying anticipated changes, biomarkers associated with specific structural and functional brain alterations will be measured in blood samples. To complement the research measures, medically relevant test results already collected by the Space Medicine Operations Division will be shared.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Detailed astronaut follow-up after return from spaceflight has historically been limited to a few weeks, which prevents conclusions about long-term health consequences of astronauts, especially after longer stays in space. This study will follow astronauts for up to 5 years after >=6 months missions. The research partially translates to similar stressful long-term exposure situations on Earth.
Task Progress:	Major accomplishments during this reporting period for this study include: • Began data collection and collected data from two crewmember participants • Began recruitment of control participants The Co-PI team is also holding bi-weekly meetings to coordinate ongoing data collection and analysis efforts for the LTH Call Back study. Ongoing forward work with Research Operations and Integration (ROI) for “LTH Call Back” includes: • The LTH Call Back study will continue recruitment and data collection, including during recruitment 2 astronauts and 4 control TSS participants during the next reporting period.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Articles in Peer-reviewed Journals	Moore TM, Basner M. "An alternative to the Bland-Altman repeated-measures correlation to account for variability of slopes across persons." Mathematics. 2025 Feb 4;13(3):512. https://doi.org/10.3390/math13030512 , Feb-2025